

NIORT SOUCHE AP, France

WMO: 073300

Lat: 46.3156N

Lon: 0.4003W

Elev: 203

StdP: 14.59

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.5	(c) 27.8	(d) 18.9	(e) 14.3	(f) 27.3	(g) 22.4	(h) 16.9	(i) 30.5	(j) 29.9	(k) 52.6	(l) 26.6	(m) 50.1	(n) 6.6	(o) 60	(p) 0.526

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 19.8	(c) 89.8	(d) 69.1	(e) 86.1	(f) 67.5	(g) 82.5	(h) 66.4	(i) 71.3	(j) 84.4	(k) 69.6	(l) 82.1	(m) 68.0	(n) 79.1	(o) 8.0	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.2	100.7	76.1	65.5	95.0	73.6	64.1	90.2	71.8	35.2	84.8	33.7	82.3	32.3	79.2	79.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.1	20.2	17.9	DB	20.8	96.5	4.2	3.0	17.8	98.7	15.3	100.5	12.9	102.2	9.9	104.4	
			WB	19.9	74.3	4.1	1.7	16.9	75.5	14.5	76.6	12.2	77.5	9.2	78.8	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.9	42.1	43.4	48.0	52.1	58.6	65.3	68.6	68.2	63.0	57.1	48.1	43.2
	DBStd	11.34	6.89	6.51	5.86	5.88	5.79	5.93	5.62	5.62	5.46	6.23	6.37	7.11
	HDD50	948	252	194	108	41	3	0	0	0	0	15	108	227
	HDD65	4099	709	605	528	387	214	67	25	25	102	252	508	677
	CDD50	2725	8	9	45	105	268	459	577	565	389	235	50	15
	CDD65	401	0	0	0	1	14	76	137	125	41	7	0	0
	CDH74	3908	0	0	1	32	179	799	1275	1205	382	35	0	0
	CDH80	1329	0	0	0	2	35	270	455	466	98	3	0	0
Wind	WSAvg	8.4	9.3	9.5	9.5	8.8	8.0	7.8	7.8	7.2	7.5	7.9	8.5	9.2
Precipitation	PrecAvg	34.9	3.6	2.9	2.4	2.8	2.7	2.2	2.2	2.2	2.6	3.7	3.9	3.9
	PrecMax	46.7	6.5	5.7	8.0	8.7	5.3	4.8	5.6	5.4	9.1	7.2	9.4	7.6
	PrecMin	21.9	0.7	0.2	0.1	0.5	0.7	0.5	0.2	0.2	0.5	0.5	1.0	0.7
	PrecStd	5.7	1.7	1.7	1.8	1.8	1.2	1.0	1.1	1.5	1.9	1.8	2.0	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.2	62.8	70.5	78.1	84.5	92.4	94.5	96.3	87.6	78.1	66.0	58.7
		MCWB	54.2	52.9	56.5	61.6	66.4	70.5	68.8	70.9	67.2	64.4	58.8	54.9
	2%	DB	55.4	57.6	65.7	72.8	78.9	86.8	89.1	89.8	82.7	73.2	62.1	56.6
		MCWB	53.0	51.0	54.3	59.2	63.8	68.1	68.3	69.8	65.1	62.7	57.5	54.1
	5%	DB	53.6	54.5	61.4	67.9	74.7	82.6	85.2	85.2	78.3	69.5	59.4	54.9
		MCWB	51.5	50.3	52.9	56.6	62.0	66.9	67.0	68.1	63.9	61.6	56.2	52.8
	10%	DB	51.8	52.6	57.6	63.6	70.6	78.1	81.2	80.7	74.4	66.2	57.3	53.3
		MCWB	49.8	49.4	51.2	54.3	60.2	65.0	65.9	66.3	62.3	60.4	54.4	51.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	54.6	58.0	63.2	68.9	73.3	72.4	74.7	69.5	66.5	60.6	56.5
		MCDB	56.3	58.4	67.1	75.3	80.9	86.2	86.6	89.8	80.0	72.8	63.0	57.5
	2%	WB	53.6	52.7	55.8	60.1	65.6	70.4	70.2	71.8	67.3	64.5	58.5	54.7
		MCDB	54.9	55.8	62.8	70.8	75.4	83.1	84.0	84.3	77.5	69.9	60.9	56.1
	5%	WB	51.9	51.4	54.1	57.5	63.0	68.4	68.7	69.7	65.5	62.8	56.6	53.1
		MCDB	53.4	53.9	59.3	66.1	72.2	79.8	81.1	82.0	74.6	67.9	59.0	54.6
	10%	WB	50.1	49.9	52.4	55.2	61.0	66.1	67.1	67.6	63.8	61.1	54.7	51.5
		MCDB	51.6	52.2	56.4	61.9	68.9	76.1	77.9	77.9	71.8	65.4	57.0	53.1
Mean Daily Temperature Range	5% DB	MDBR	10.6	13.5	16.3	18.2	18.4	19.7	19.8	20.4	19.5	15.5	12.3	11.0
		MCDBR	10.9	16.2	22.3	25.1	25.0	26.7	26.0	27.8	25.3	19.1	13.3	11.1
	5% WB	MCWBR	9.0	11.3	13.3	13.9	12.7	11.4	9.9	11.1	11.2	10.4	9.5	9.0
		MCWBR	9.8	14.0	18.0	22.9	22.4	24.1	22.2	24.1	21.3	16.0	12.4	10.3
Clear-Sky Solar Irradiance	taub		0.325	0.341	0.371	0.395	0.399	0.412	0.402	0.392	0.366	0.364	0.338	0.321
		taud	2.444	2.389	2.307	2.240	2.282	2.280	2.305	2.352	2.401	2.418	2.436	2.433
	Ebn at Noon		244	262	270	273	275	271	272	271	269	250	235	231
		Edh at Noon	22	28	35	41	41	41	40	36	32	27	22	20
All-Sky Solar Radiation	RadAvg		379	679	1053	1495	1755	1973	1906	1686	1335	799	459	334
	RadStd		38	70	117	149	138	159	96	125	87	68	49	41

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page